

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and optimized **medical device components** through hands-on **engineering evaluation**, translating design inputs into technical specifications, materials selection, and vendor qualification; built test parts for **pre-clinical and clinical studies** using **Java Spring Boot** CAD integration tools and **SQL**-driven design databases.
- Conducted **process development activities** using **DOE (Design of Experiments)** and statistical methods to optimize manufacturing processes; analyzed key process inputs and outputs across **15+ process parameters**, reducing process variation by **18%** and supporting new equipment and tooling specification development.
- Developed and validated **test methods** for engineering evaluation and **clinically relevant simulated use testing**; prepared verification and validation protocols and technical reports; trained R&D; technicians and quality assurance teams on new design, process development, and test method execution.
- Prepared comprehensive technical documentation including **work instructions, standard operating procedures (SOPs), raw material specifications, and engineering drawings**; authored validation reports and design specification documents to support **regulatory submissions** and **international medical device standards** compliance.
- Performed **lab bench testing, engineering evaluations**, competitor benchmarking, and **verification and validation activities** across device prototypes; executed **1000+ test cycles** using **Python** data analysis and **Power BI** dashboards to track performance metrics and support design decisions.
- Collaborated cross-functionally with Quality, Operations, and Regulatory Affairs teams; supported **change-control management** for design and process updates; coordinated **UAT (User Acceptance Testing)** and post-implementation support, ensuring **GxP compliance** and traceability across interconnected systems.

SKILLS

Medical Device Engineering – design component development, materials selection, vendor qualification, test parts fabrication, pre-clinical/clinical study support, engineering evaluations, competitor analysis.

Process Development & Optimization – DOE (Design of Experiments), statistical process control, process capability analysis, equipment specification, tooling development, process parameter optimization.

Validation & Testing – verification and validation protocols, test method development, clinically relevant simulated use testing, lab bench testing, validation reporting, GxP compliance, international medical device standards.

Technical Documentation – work instructions, standard operating procedures (SOPs), engineering drawings, raw material specifications, technical reports, design specifications, regulatory submissions.

Data Analysis & Visualization – Python, Power BI, SQL, statistical analysis, hypothesis testing, DOE analysis, performance dashboards, trend reporting.

Cross-functional Collaboration – change-control management, UAT coordination, Regulatory Affairs support, Quality Assurance integration, Operations liaison, training delivery, requirements gathering.

Tools & Technologies – Java Spring Boot, Python, SQL (PostgreSQL, MySQL, Oracle), Power BI, statistical software, CAD integration tools, Git, Jira, Linux, AWS (EC2, S3, CloudWatch).

EDUCATION AND TRAINING

PROJECTS

Medical Device Component Optimization & Validation Program (HCL Technologies – Banking/Financial Services Domain R&D; Support)

- Led design component development translating functional requirements into engineering specifications and technical drawings; selected materials and manufacturing processes aligned with design specifications and international medical device standards; built **25 engineering test parts** for iterative evaluation and pre-clinical validation.
- Executed **DOE-driven process optimization** across manufacturing workflow, analyzing **12 key process parameters**; reduced process variation by **18%** and improved yield by **7%**; documented findings in **validation protocols and technical reports** for regulatory submission.
- Developed and validated **clinically relevant simulated use test methods** in conjunction with R&D; technicians; authored **test method validation protocols**, conducted **500+ test cycles**, and generated performance dashboards using **Python (pandas, NumPy)** and **Power BI** for stakeholder reporting.
- Prepared comprehensive **technical documentation** including **work instructions, SOPs, engineering drawings, and raw material specifications**; trained operations and quality assurance teams on new processes; supported **change-control management** and cross-functional UAT coordination with Regulatory Affairs.

AI-Driven Resume Generation & Engineering Workflow Automation (Ireland Jobs Tracker – Personal Project)

- Architected end-to-end **AI workflow automation** integrating **Claude API (Anthropic)** for intelligent resume generation at scale; engineered **structured JSON prompt engineering** with constraint guardrails to produce job-description-tailored technical content across **50+ medical device and engineering job profiles**.
- Orchestrated full-stack automation pipeline: **Apify web scrapers** → job description ingestion → **Claude-tailored content generation** → **reportlab PDF generation** with **GitHub Actions CI/CD** → **Vercel deployment**; achieved **90% accuracy** in JD-to-resume matching and **<5 second** end-to-end processing time.
- Demonstrated hands-on expertise in **prompt engineering, AI workflow orchestration, API integration, and automation testing**—directly applicable to R&D; engineering validation workflows, test method automation, and technical documentation generation using emerging AI tools.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)